1.5.1 Properties of Division

Can you find the value when (–5) is divided by 3? The value is not an integer. Clearly, the collection of integers is not 'closed' under division. Test it by taking 5 more examples

To test the commutative property of division on integers, let us take –5 and 3.

$(-5) \div 3 \neq 3 \div (-5)$. Also $5 \div (-3) \neq (-3) \div 5$ and $(-5) \div (-3) \neq (-3) \div (-5)$

Therefore, division of integers are not commutative. Verify this with 5 more examples.

Since the commutative property is not true, associative property does not hold.

Let us take integers, –7 and 1.

$(-7) \div 1 = -7$ but $1 \div (-7) \neq -7$.

Let us take $(-6) \div (-1)$ and $6 \div (-1)$.

$(-6) \div (-1) = 6$ and $6 \div (-1) = -6$.

Therefore, when we divide an integer by –1, we will not get the same integer.

Hence, there does not exist an identity for division of integers.

Take any five integers and verify the above.

Note

An integer divided by zero is not defined. But zero divided by a non-zero integer is zero.

Exercise 1.4

1. Fill in the blanks.
   (i) $(-40) \div ____ = 40$ (ii) $25 \div ____ = -5$
   (iii) $____ \div (-4) = 9$ (iv) $(-62) \div (-62) = ____$
2. Say true or false.
   (i) $(-30) \div (-6) = -6$ (ii) $(-64) \div (-64)$ is 0
3. Find the values of the following.
   (i) $(-75) \div 5$ (ii) $(-100) \div (-20)$
   (iii) $45 \div (-9)$ (iv) $(-82) \div 82$
4. The product of two integers is −135. If one number is −15, Find the other integer.
5. In 8 hours duration, with uniform decrease in temperature, the temperature dropped 24°. How many degrees did the temperature drop each hour?



6. An elevator descends into a mine shaft at the rate of 5 m/min. If the descent starts from 15 m above the ground level, how long will it take to reach –250 m?
7. A person lost 4800 calories in 30 days. If the calory loss is uniform, calculate the loss of calory per day.
8. Given $168 \times 32 = 5376$ then, find $(-5376) \div (-32)$.

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